Just IT

B2W group

Apprenticeships | Training | Recruitment

Data Technician

Name: Jonathan Akpofure

Course Date:

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Day 1: Task 1

Please research and complete the below questions relating to key concepts of databases.

What is a primary key?	A primary key is a set of columns used to uniquely identify every record in a database
How does this differ from a secondary key?	A primary key ensures each row is unique whilst a secondary key improves the efficiency of queries that do not use the primary key
How are primary and foreign keys related?	Foreign key column in a table normally means there is a column with unique values in another table. Foreign keys are used to link data in one table to data in another table
Provide a real-world example of a one-to-one relationship	Marriage- one person is legally married to only one person at a time
Provide a real-world example of a one-to-many relationship	Department to employees- one department has many employees but each employee belongs to only one department
Provide a real-world example of a many-to-many relationship	Author and books- An author can write many books, and a book can have many authors Students and classes- One student can have many classes, and a class can have many students.

Day 1: Task 2

Please research and complete the below questions relating to key concepts of databases.

What is the difference between a relational and non-relational database?	Relational databases store structural data in tables with rows columns showing the relationships whilst non-relational databases use flexible models for large amounts of data. Non-relational is better for unstructured data
What type of data would benefit off the non-relational model? **Why?**	Unstructured data- data that doesn't have a structure which makes it hard to store, process and analyse such as emails and videos

Day 3: Task 1

Please research the below 'JOIN' types, explain what they are and provide an example of the types of data it would be used on.

Self-join	It's a type of join query used to compare rows within the same table Categorical Data
Right join	It's a type of join query that retrieves data from the right table and the matching data from the left table Compatible/Relational data so you can merge the data from the tables
Full join	It's a type of join query that returns all rows from both tables; filling in NULL for missing matches Merging Data
Inner join	It's a type of join query that returns only the matching rows between both tables Relational Data
Cross join	It's a type of database join that combines every row from one table with every row from another table

	Qualitative Data
Left join	It's a type of join query that returns all rows from the right table and matching rows from the right table Compatible/Relational Data

Day 4: Task 1: Written

In your groups, discuss and complete the below activity. You can either nominate one writer or split the elements between you. Everyone however must have the completed work below:

Imagine you have been hired by a small retail business that wants to streamline its operations by creating a new database system. This database will be used to manage inventory, sales, and customer information. The business is a small corner shop that sells a range of groceries and domestic products. It might help to picture your local convenience store and think of what they sell. They also have a loyalty program, which you will need to consider when deciding what tables to create.

Write a 500-word essay explaining the steps you would take to set up and create this database. Your essay should cover the following points:

1. ***Understanding the Business Requirements**:*
    a. *What kind of data will the database need to store?*
    b. *Who will be the users of the database, and what will they need to accomplish?*
2. ***Designing the Database Schema**:*
    a. *How would you structure the database tables to efficiently store inventory, sales, and customer information?*
    b. *What relationships between tables are necessary (e.g., how sales relate to inventory and customers)?*
3. ***Implementing the Database**:*
    a. *What SQL commands would you use to create the database and its tables?*
    b. *Provide examples of SQL statements for creating tables and defining relationships between them.*
4. ***Populating the Database**:*
    a. *How would you input initial data into the database? Give examples of SQL INSERT statements.*
5. ***Maintaining the Database**:*
    a. *What measures would you take to ensure the database remains accurate and up to date?*
    b. *How would you handle backups and data security?*

Your essay should include specific examples of SQL commands and explain why each step is necessary for creating a functional and efficient database for the retail business.

Please write your 500-word essay here

To set up and create the database, it is imperative to understand the business requirements.

The database needs to store structured data. This data includes some table names and their corresponding attributes. The table names are as follows: Customer, Product, Transaction, and Sales Items Tables.

The Customer Table includes Customer ID (Primary Key), Customer names (first and last names), Loyalty Points, and contact details (phone number). The Product Table includes the Product ID (Primary Key), Product Name, Product Category, Price, and Stock Quantity. The Transaction Table includes the Transaction ID (Primary Key), Transaction Total (amount spent), Transaction Date, and Customer ID. Lastly, Sales Items Table includes Sales Item ID (Primary Key), Sales ID (Foreign Key), Product ID (Foreign Key), Quantity and Price

The database has specific users and tasks related to them. Business Owners/managers can use the database to track and manage cash flow to make correct business decisions. They would also have access to personal data from customers and all the products sold within the shop. Customers can request personal data from the shop, such as duplicate receipts, and locate specific products. External Parties, such as Accountants and Financiers, would have access to sales records so that they can generate reports based on revenue and track sales by date and product to help identify patterns.

The next step is to understand how the table names are linked to each other to help create the database. For example, we know that the transaction table links to the customer table because there is a customer ID as an attribute in both.

Subsequently, the next step is to implement the database. We would need to use an SQL Command to create the table, which would be CREATE DATABASE Convenience_Store_1 ;

To create the tables, we would need multiple SQL statements, which are as follows:

```
CREATE TABLE Customers (
   CustomerID INT AUTO_INCREMENT PRIMARY KEY,
   First_Name VARCHAR (50) NOT NULL,
   Last_Name VARCHAR (50) NOT NULL,
```

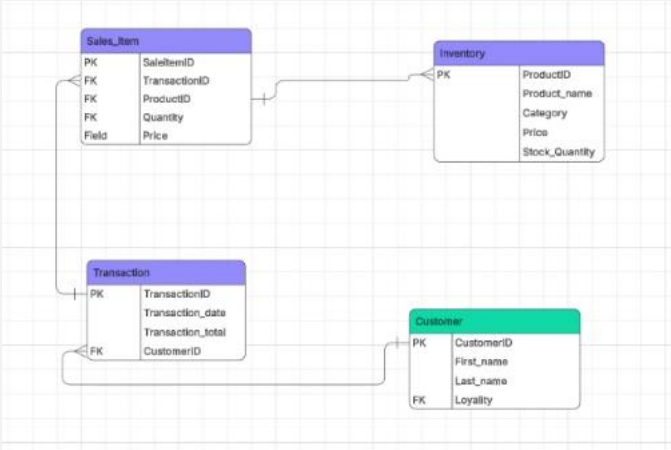
```
  Phone VARCHAR (20),
  Loyalty_Points INT DEFAULT

CREATE TABLE Inventory (
  ProductID INT AUTO_INCREMENT PRIMARY KEY,
  Product_Name VARCHAR (100) NOT NULL,
  Category VARCHAR (50),
  Price DECIMAL (8,2) NOT NULL,
  Stock_Quantity INT NOT NULL


CREATE TABLE Transactions (
  TransactionID INT AUTO_INCREMENT PRIMARY KEY,
  Sale_Date DATETIME NOT NULL,
  Sale_Total DECIMAL (10,2) NOT NULL,
  CustomerID INT,
  FOREIGN KEY (CustomerID) REFERENCES Customers (CustomerID)


CREATE TABLE SalesItems (
  SaleItemID INT AUTO_INCREMENT PRIMARY KEY,
  TransactionID INT NOT NULL,
  ProductID INT NOT NULL,
  Quantity INT NOT NULL,
  Price DECIMAL (8,2) NOT NULL,


FOREIGN KEY (TransactionID) REFERENCES Transactions
(TransactionID),
  FOREIGN KEY (ProductID) REFERENCES Inventory (ProductID
```



The next step is to obtain the initial data present in the database using specific SQL Insert statements.

These are as followed

```
INSERT INTO Products (ProductID, ProductName, Category, Price, StockQuantity)
VALUES (1, 'Milk', 'Dairy', 1.50, 50);
```

```
INSERT INTO Customers (CustomerID, Name, Email, LoyaltyPoints)
VALUES (1, 'Alice Wong, ’ 'alice@example.com', 10);
```

```
INSERT INTO Sales (SaleID, SaleDate, CustomerID)
VALUES (1, '2026-02-05', 1);
```

```
INSERT INTO SalesDetails (SaleDetailID, SaleID, ProductID, Quantity, Price)
VALUES (1, 1, 1, 2, 3.00);
```

Arguably, the most important part is maintaining the database by implementing certain measures. To ensure accuracy we would need to regularly update stock quantities after each sale and perform audits to match inventory with physical stock

Schedule daily, weekly, and monthly backups. You would also need to use access controls so that only authorized members of the staff can see and modify the data. Encrypting sensitive data is also necessary to make it unreadable to unauthorized members.

Day 4: Task 2: SQL Practical

In your groups, work together to answer the below questions. It may be of benefit if one of you shares your screen with the group and as a team answer / take screen shots from there.

Setting up the database:

1. **Download world_db(1)**
2. **Follow each step to create your database**

For each question I would like to see both the syntax used and the output.

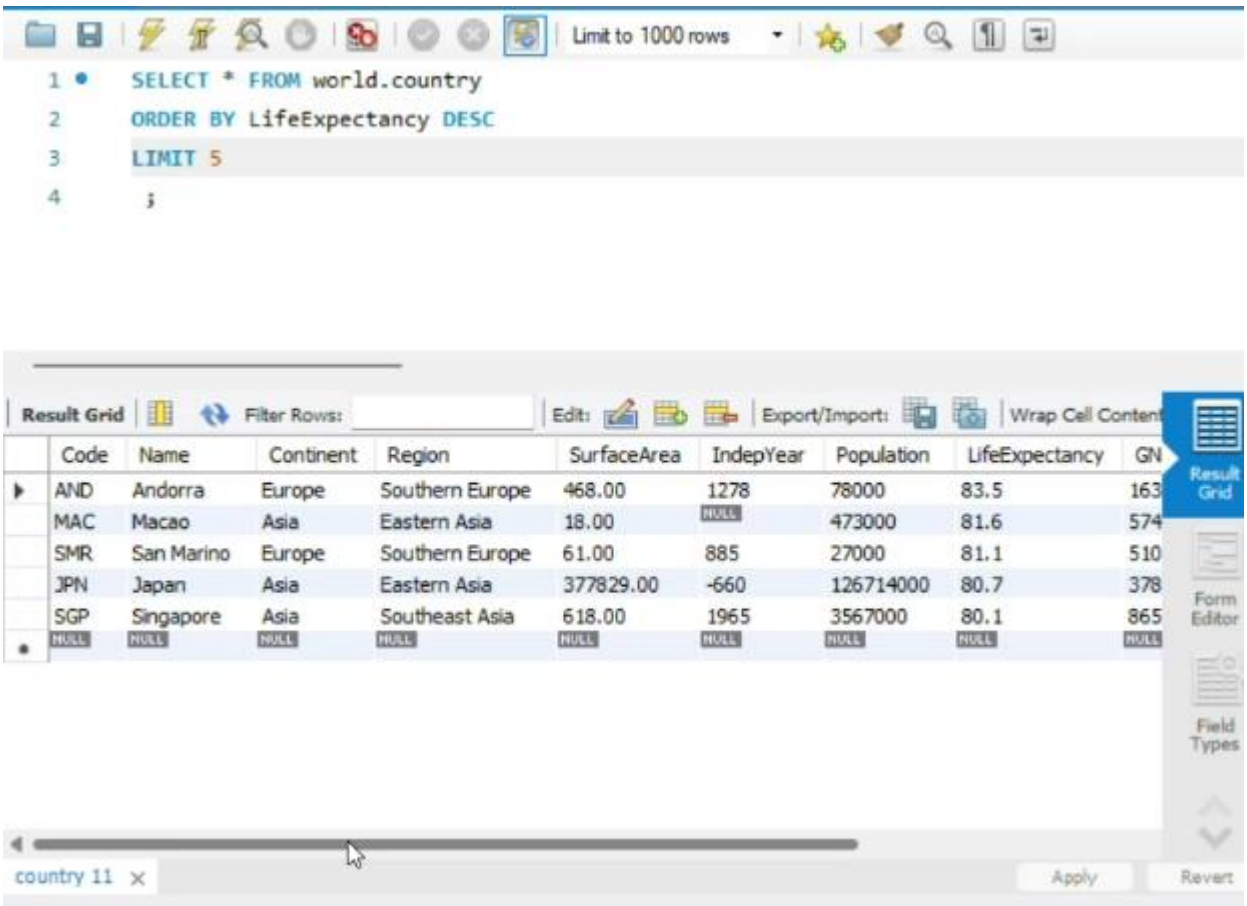
1. **Count Cities in USA:** *Scenario:* You've been tasked with conducting a demographic analysis of cities in the United States. Your first step is to determine the total number of cities within the country to provide a baseline for further analysis.

```
SELECT * FROM city
WHERE country_id = 103
LIMIT 5
;
```

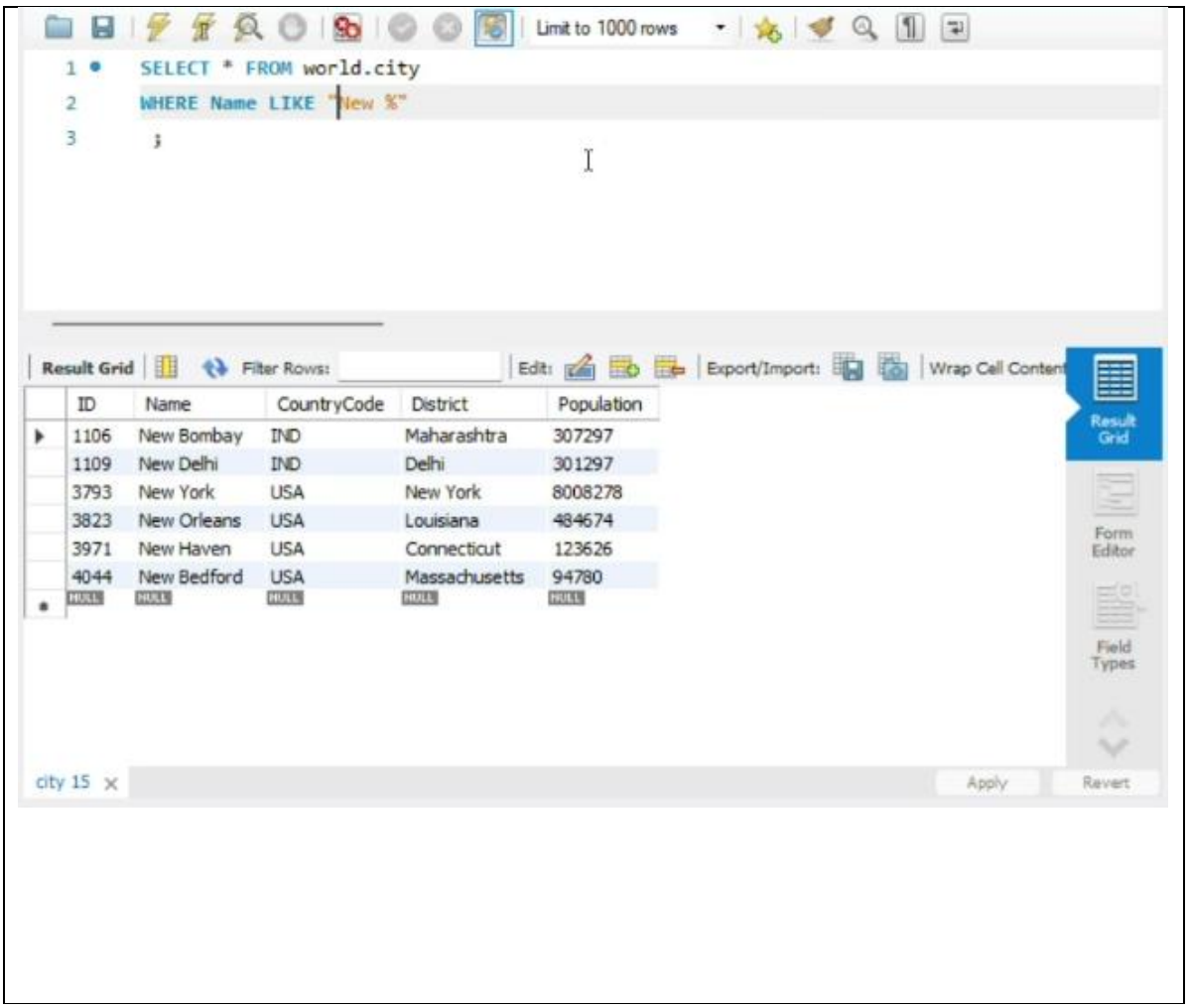
Result Grid | Filter Rows: | Edit: | Exp

city_id	city	country_id	last_update
11	Akron	103	2006-02-15 04:45:25
33	Arlington	103	2006-02-15 04:45:25
41	Augusta-Richmond County	103	2006-02-15 04:45:25
42	Aurora	103	2006-02-15 04:45:25
65	Bellevue	103	2006-02-15 04:45:25
NULL	NULL	NULL	NULL

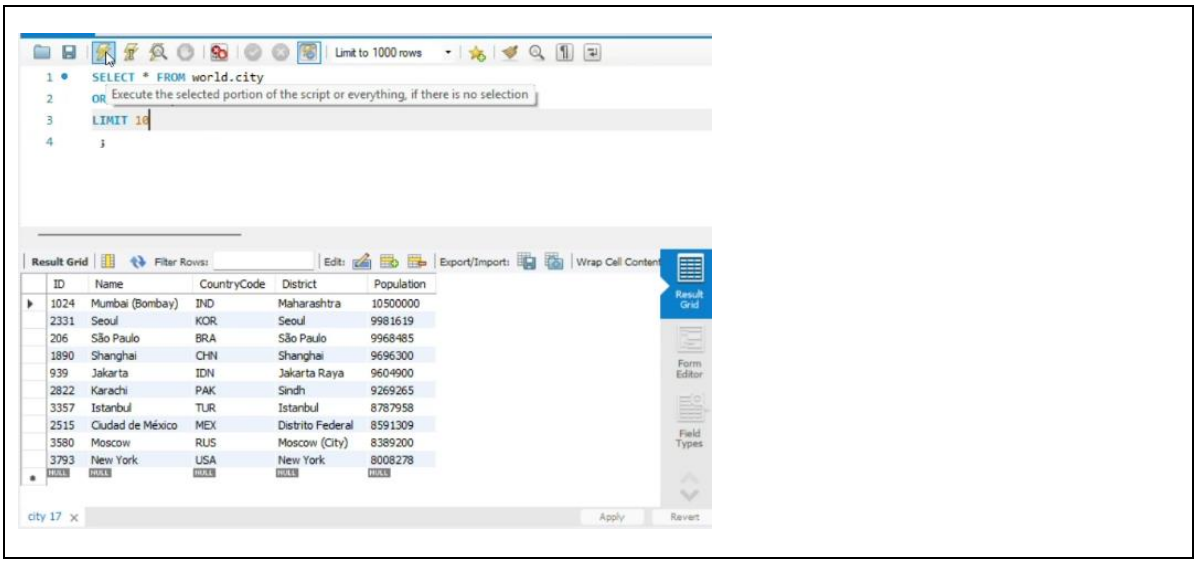
2. **Country with Highest Life Expectancy:** *Scenario:* As part of a global health initiative, you've been assigned to identify the country with the highest life expectancy. This information will be crucial for prioritising healthcare resources and interventions.



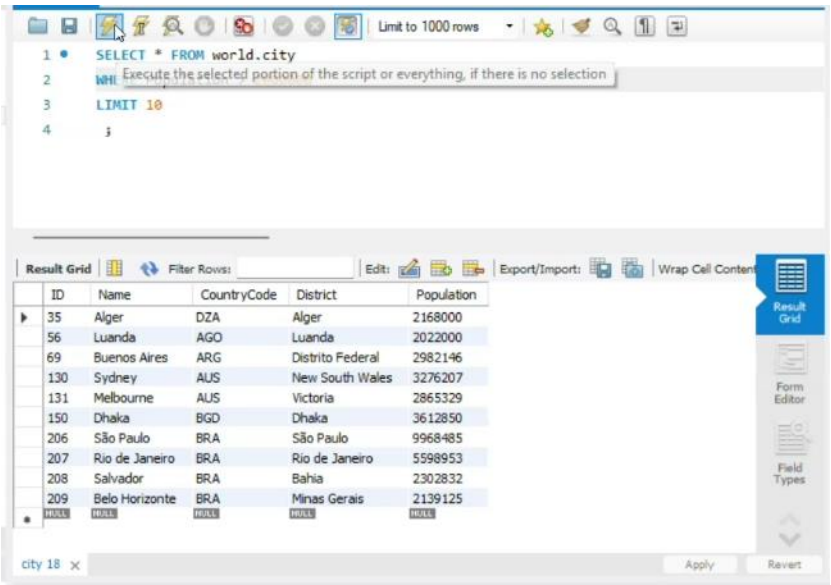
3. **"New Year Promotion: Featuring Cities with 'New :** *Scenario:* In anticipation of the upcoming New Year, your travel agency is gearing up for a special promotion featuring cities with names including the word 'New'. You're tasked with swiftly compiling a list of all cities from around the world. This curated selection will be essential in creating promotional materials and enticing travellers with exciting destinations to kick off the New Year in style.



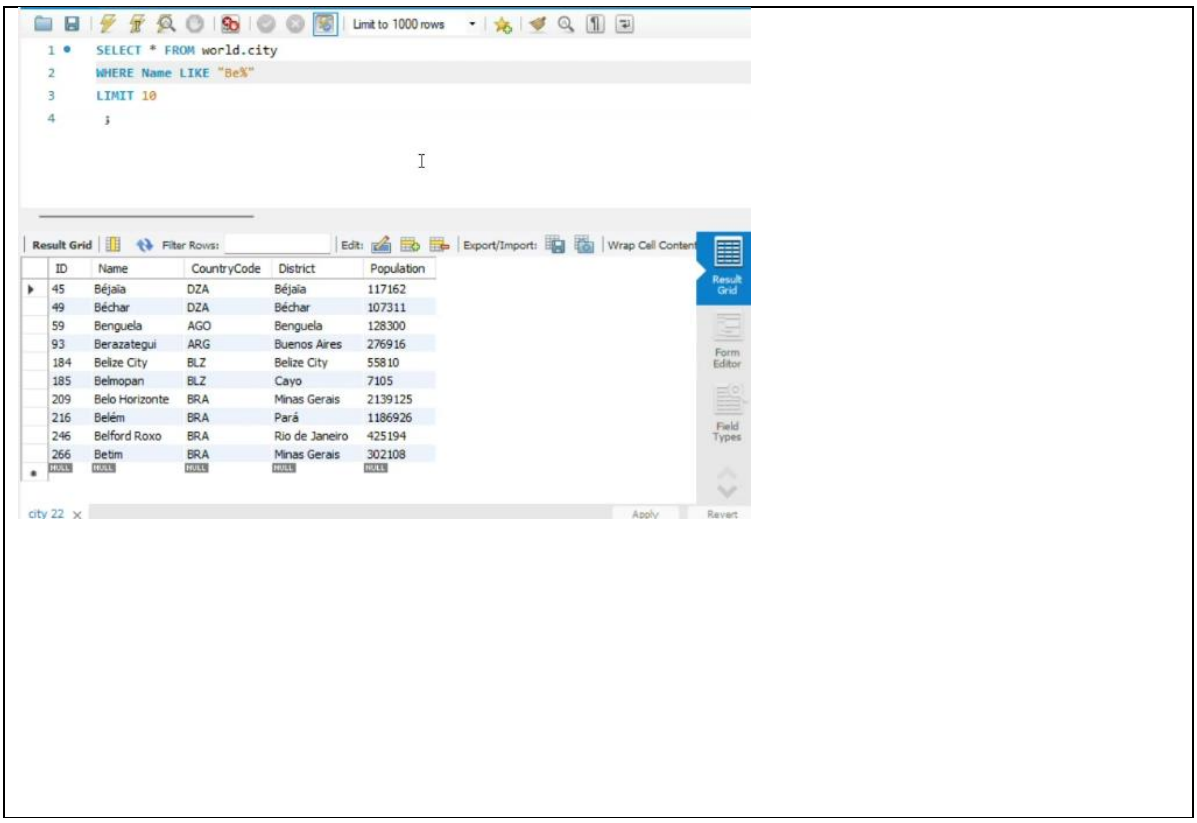
4. **Display Columns with Limit (First 10 Rows):** *Scenario:* You're tasked with providing a brief overview of the most populous cities in the world. To keep the report concise, you're instructed to list only the first 10 cities by population from the database.



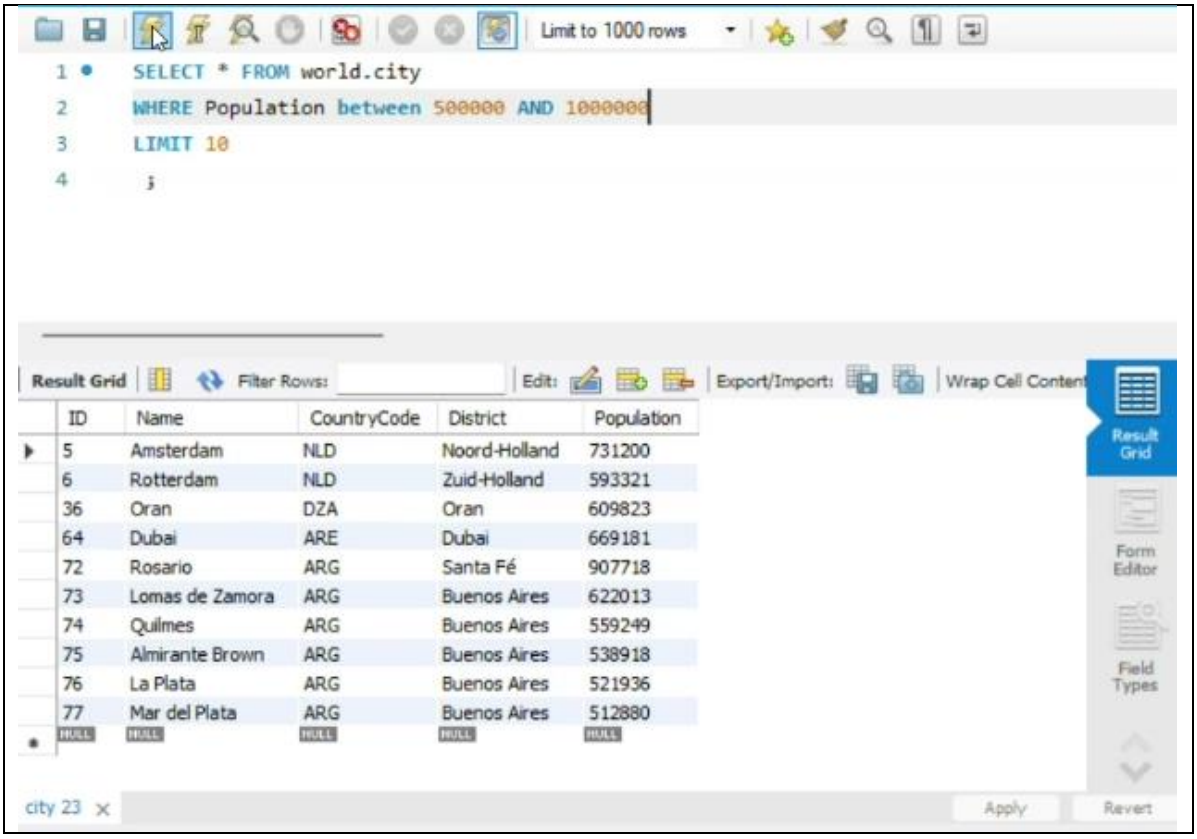
5. **Cities with Population Larger than 2,000,000:** *Scenario:* A real estate developer is interested in cities with substantial population sizes for potential investment opportunities. You're tasked with identifying cities from the database with populations exceeding 2 million to focus their research efforts.



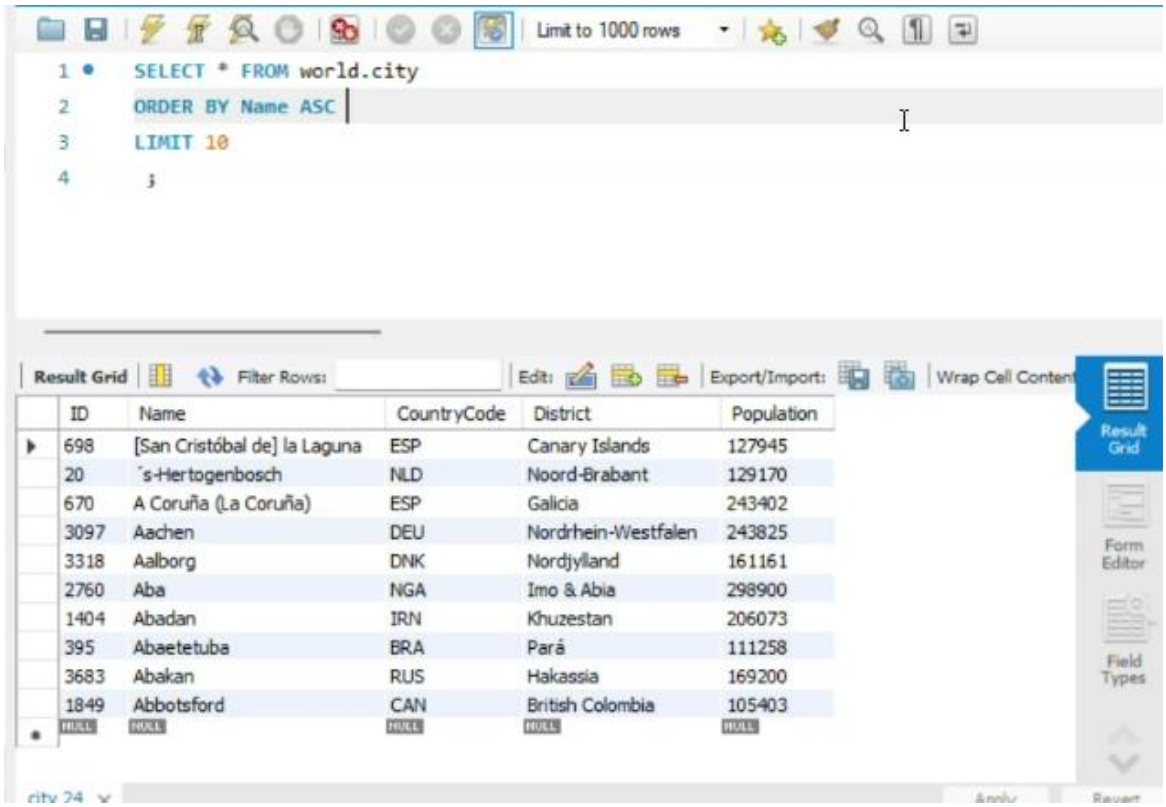
6. **Cities Beginning with 'Be' Prefix:** *Scenario:* A travel blogger is planning a series of articles featuring cities with unique names. You're tasked with compiling a list of cities from the database that start with the prefix 'Be' to assist in the blogger's content creation process.



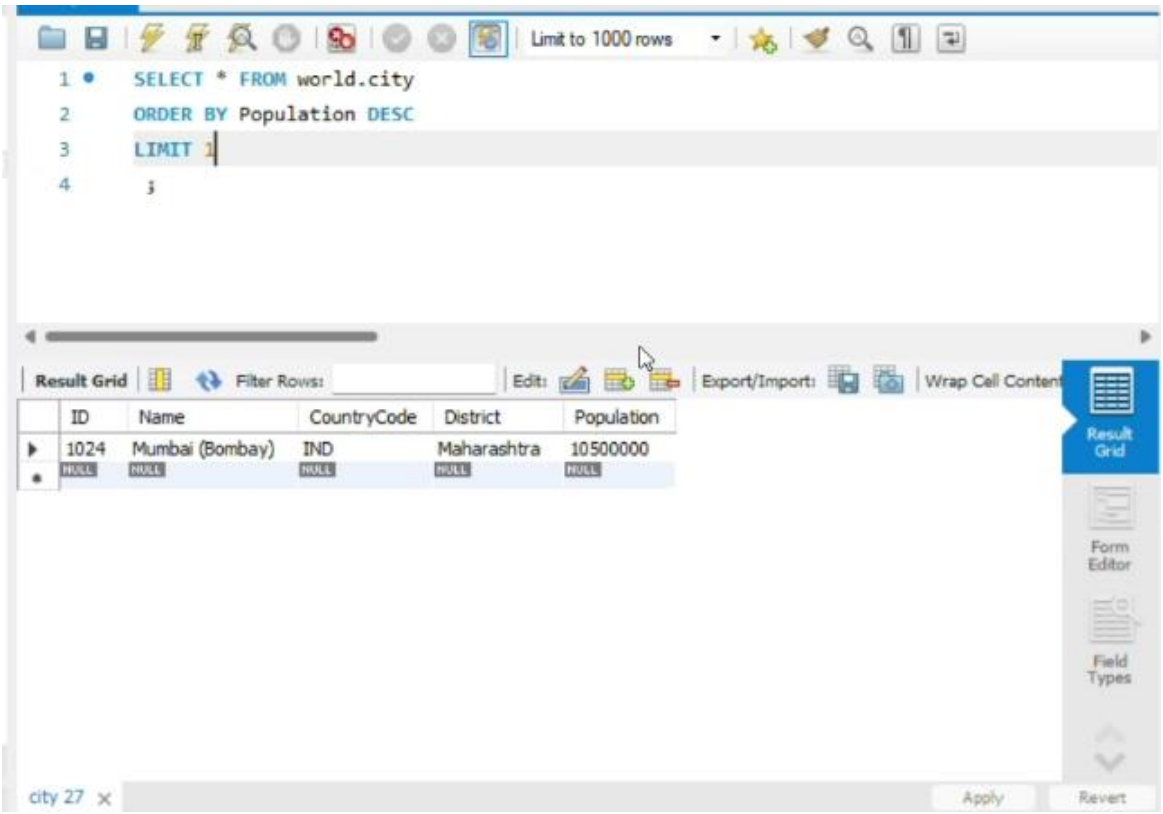
7. **Cities with Population Between 500,000-1,000,000:** *Scenario:* An urban planning committee needs to identify mid-sized cities suitable for infrastructure development projects. You're tasked with identifying cities with populations ranging between 500,000 and 1 million to inform their decision-making process.



8. **Display Cities Sorted by Name in Ascending Order:** *Scenario:* A geography teacher is preparing a lesson on alphabetical order using city names. You're tasked with providing a sorted list of cities from the database in ascending order by name to support the lesson plan.

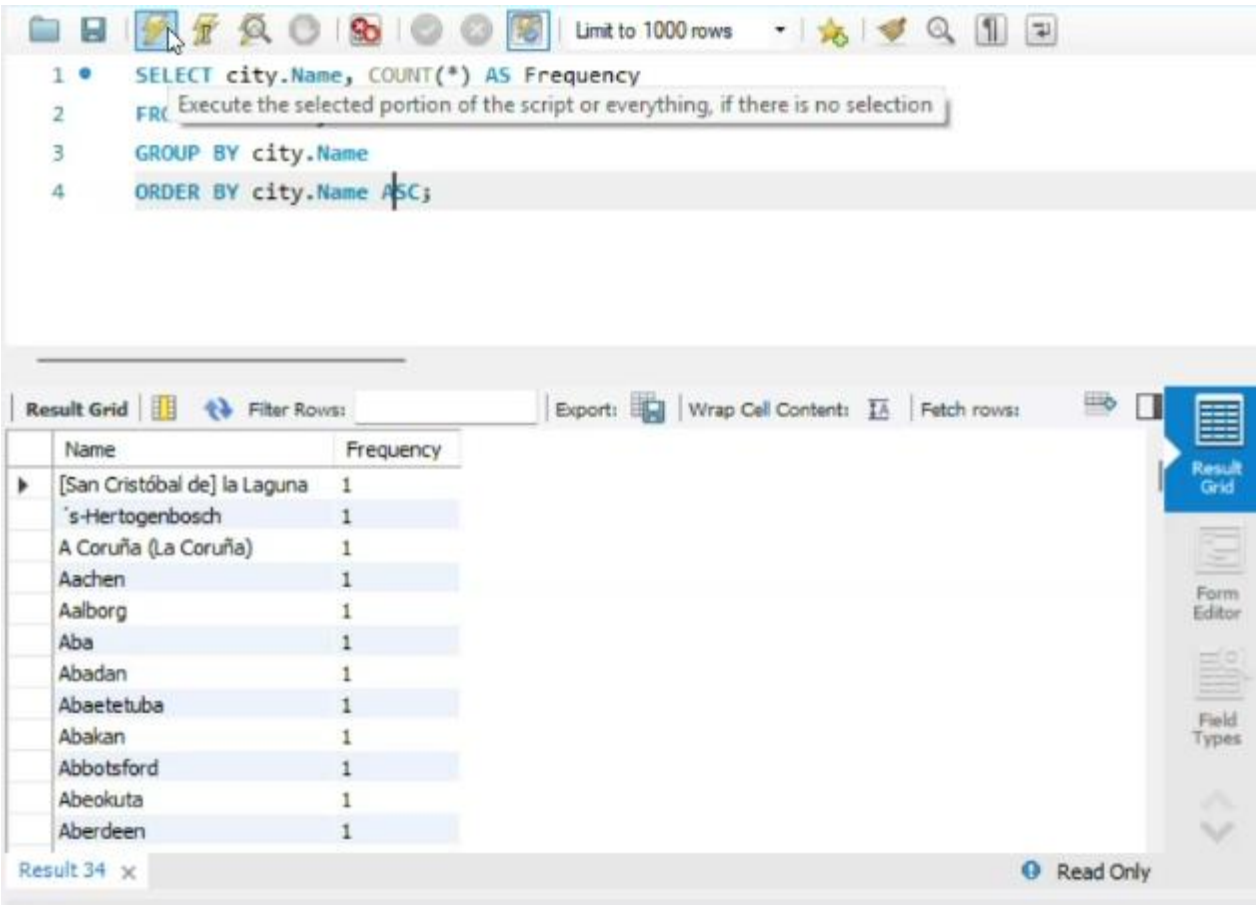


9. **Most Populated City:** *Scenario:* A real estate investment firm is interested in cities with significant population densities for potential development projects. You're tasked with identifying the most populated city from the database to guide their investment decisions and strategic planning.

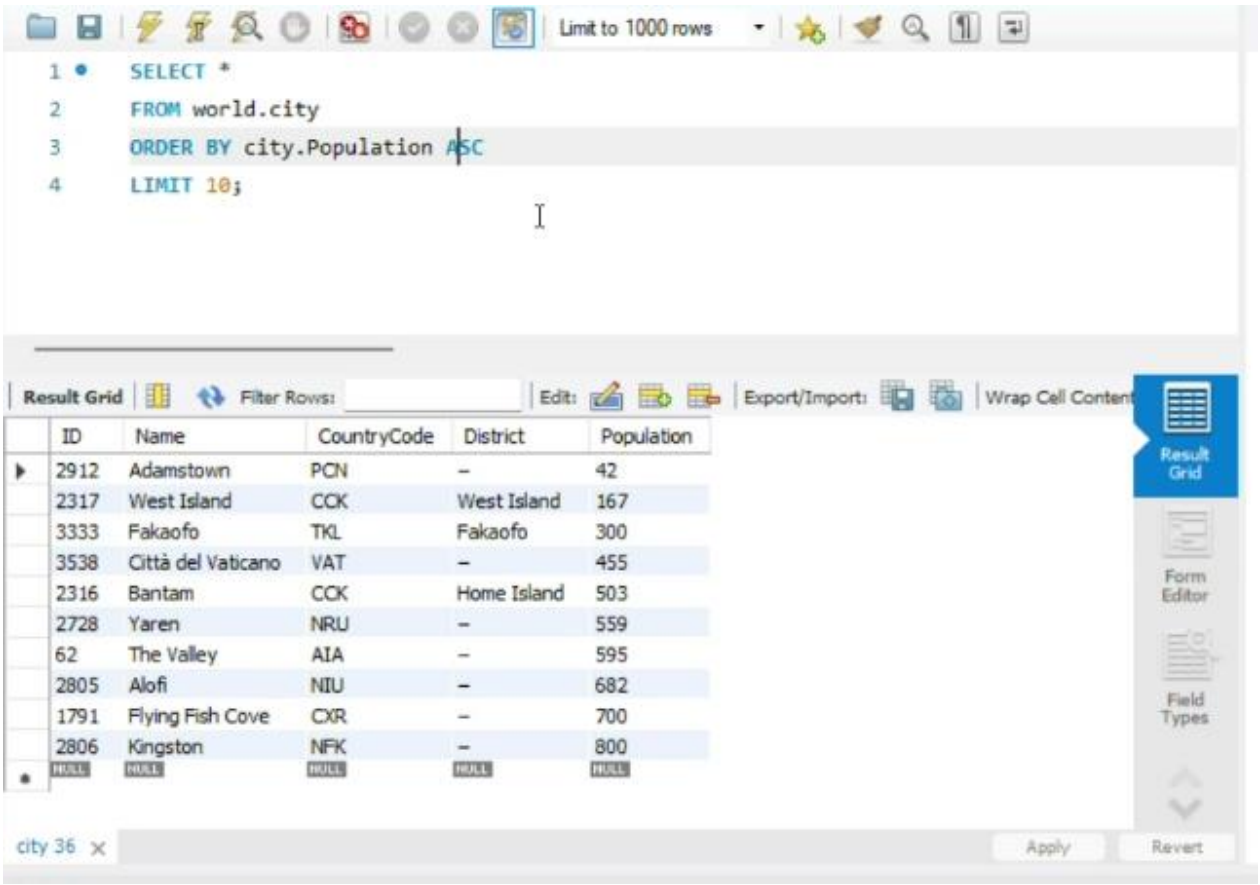


10. **City Name Frequency Analysis: Supporting Geography Education** *Scenario*: In a geography class, students are learning about the distribution of city names around the world. The teacher, in preparation for a lesson on city name frequencies, wants to provide students with a list of unique city names sorted alphabetically, along with their respective

counts of occurrences in the database. You're tasked with this sorted list to support the geography teacher.



11. **City with the Lowest Population:** *Scenario:* A census bureau is conducting an analysis of urban population distribution. You're tasked with identifying the city with the lowest population from the database to provide a comprehensive overview of demographic trends.

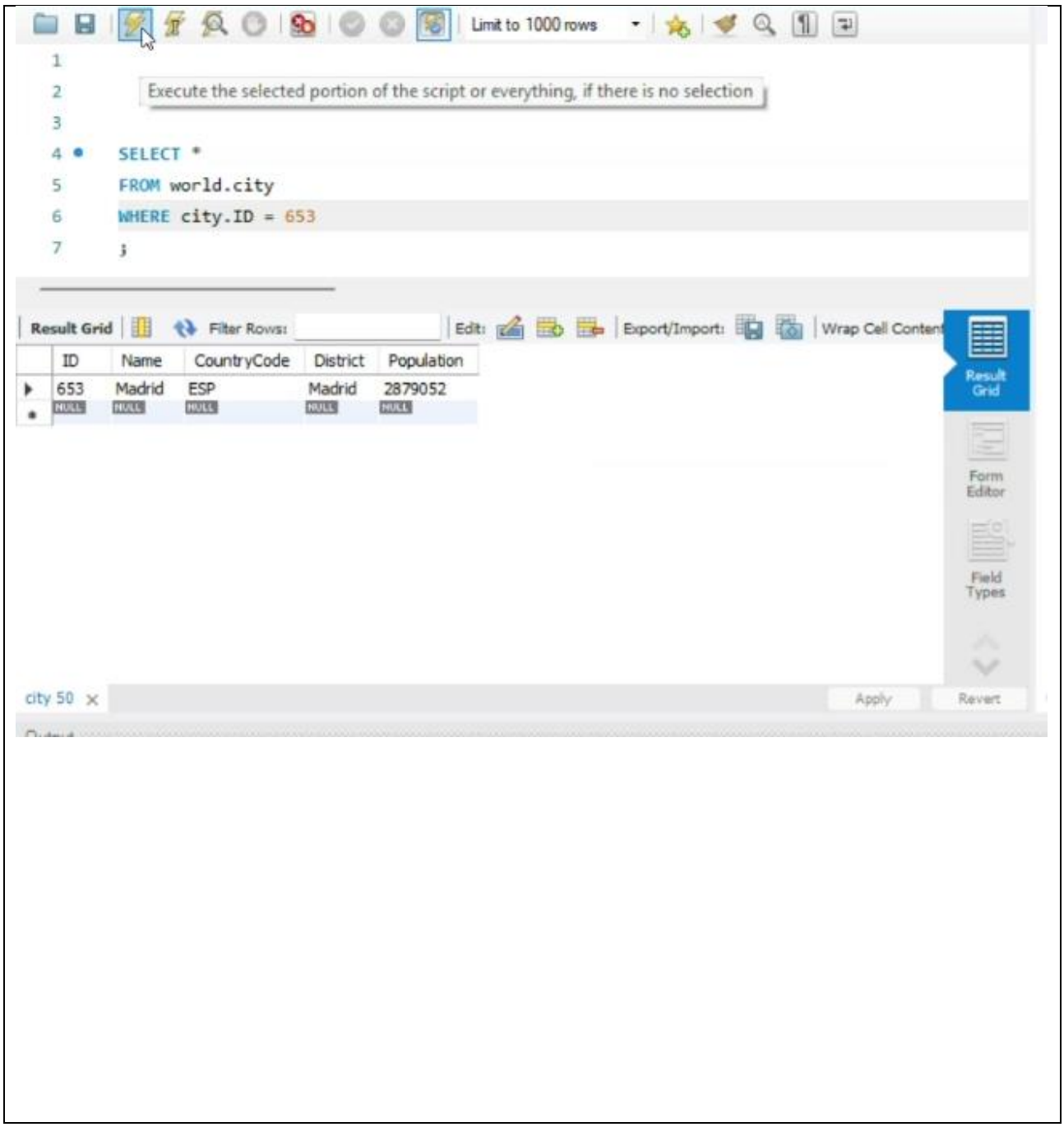


12. **Country with Largest Population:** *Scenario:* A global economic research institute requires data on countries with the largest populations for a comprehensive analysis. You're tasked with identifying the country with the highest population from the database to provide valuable insights into demographic trends.

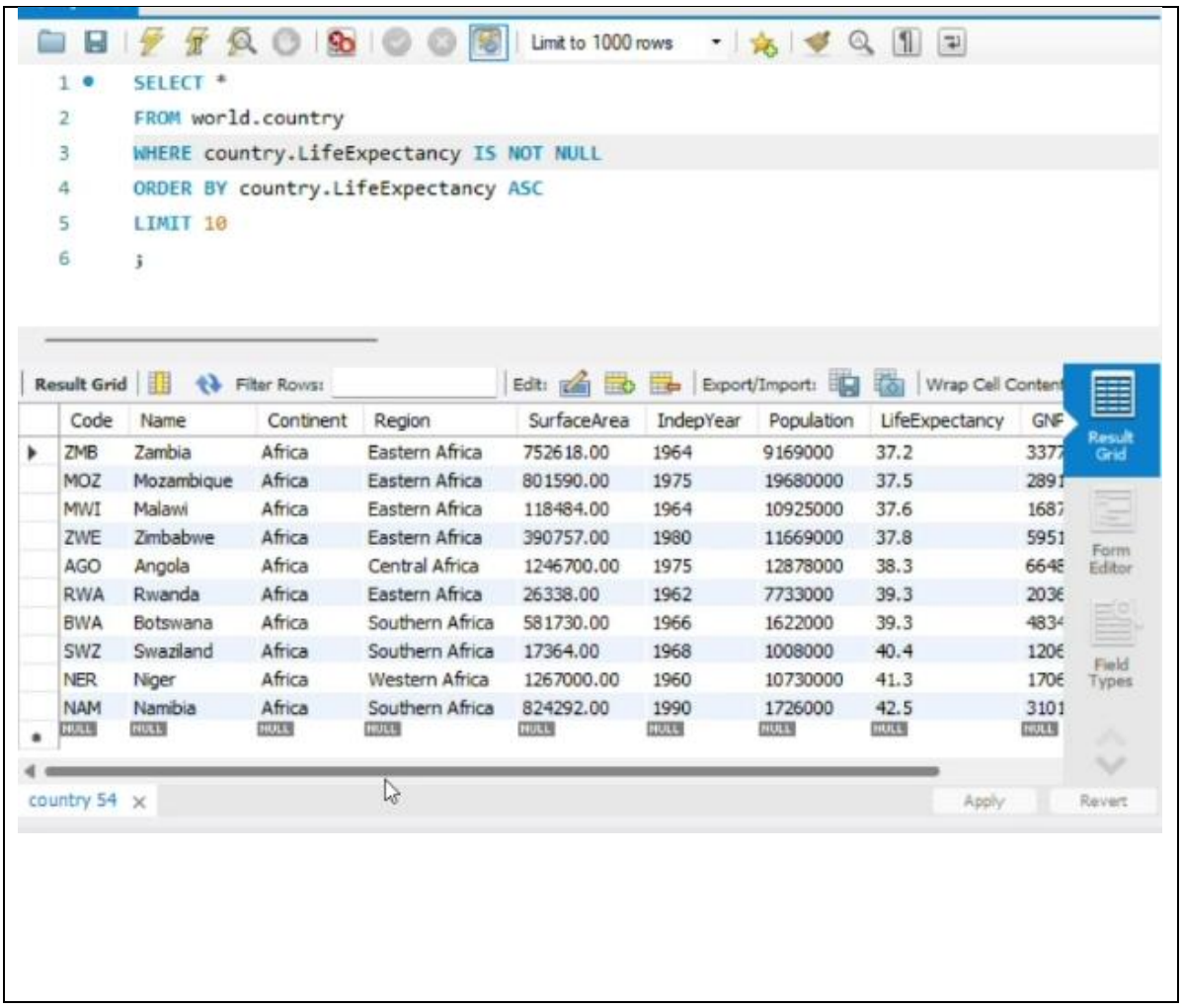
```
SELECT *
FROM world.country
ORDER BY country.Population DESC
LIMIT 10;
```

Code	Name	Continent	Region	SurfaceArea	IndepYear	Population
CHN	China	Asia	Eastern Asia	9572900.00	-1523	1277558000
IND	India	Asia	Southern and Central Asia	3287263.00	1947	1013662000
USA	United States	North America	North America	9363520.00	1776	278357000
IDN	Indonesia	Asia	Southeast Asia	1904569.00	1945	212107000
BRA	Brazil	South America	South America	8547403.00	1822	170115000
PAK	Pakistan	Asia	Southern and Central Asia	796095.00	1947	156483000
RUS	Russian Federation	Europe	Eastern Europe	17075400.00	1991	146934000
BGD	Bangladesh	Asia	Southern and Central Asia	143998.00	1971	129155000
JPN	Japan	Asia	Eastern Asia	377829.00	-660	126714000
NGA	Nigeria	Africa	Western Africa	923768.00	1960	111506000

13. **Capital of Spain:** *Scenario:* A travel agency is organising tours across Europe and needs accurate information on capital cities. You're tasked with identifying the capital of Spain from the database to ensure itinerary accuracy and provide travellers with essential destination information.



14. **Country with Shortest Life Expectancy:** *Scenario:* A healthcare foundation is conducting research on global health indicators. You're tasked with identifying the country with the highest life expectancy from the database to inform their efforts in improving healthcare systems and policies.



15. **Cities in Europe:** *Scenario:* A European cultural exchange program is seeking to connect students with cities across the continent. You're tasked with compiling a list of cities located in Europe from the database to facilitate program planning and student engagement.

```sql
SELECT DISTINCT city.Name
FROM city
JOIN country
ON city.CountryCode = country.Code
WHERE country.Continent = 'Europe'
ORDER BY city.Name ASC;
```

Result Grid

Name
[San Cristóbal de] la Laguna
´s-Hertogenbosch
A Coruña (La Coruña)
Aachen
Aalborg
Abakan
Aberdeen
Aix-en-Provence
Albacete
Alcalá de Henares
Alcorcón
Alessandria
Algeciras
Alicante [Alacant]
Alkmaar
Almere
Almería
Almetjevsk
Altševsk
Amadora
Amersfoort
Amiens
Amsterdam
Ancona
Andorra la Vella

16. **Average Population by Country:** *Scenario:* A demographic research team is conducting a comparative analysis of population distributions across countries.

You're tasked with calculating the average population for each country from the database to provide valuable insights into global population trends.

```
SELECT country.Name AS Country,
       AVG(city.Population) AS Average_Population
FROM city
JOIN country
ON city.CountryCode = country.Code
GROUP BY country.Name
ORDER BY country. Name ASC;
```

Country	Average_Population
Afghanistan	583025.0000
Albania	270000.0000
Algeria	288454.3889
American Samoa	3761.5000
Andorra	21189.0000
Angola	512320.0000
Anguilla	778.0000
Antigua and Barbuda	24000.0000
Argentina	350816.8947
Armenia	544366.6667
Aruba	29034.0000
Australia	808119.0000
Austria	397378.8333
Azerbaijan	616000.0000
Bahamas	172000.0000
Bahrain	148000.0000
Bangladesh	357079.4167
Barbados	6070.0000
Belarus	296312.5000
Belgium	178813.5556
Belize	31457.5000

17. **Capital Cities Population Comparison:** *Scenario:* A statistical analysis firm is examining population distributions between capital cities worldwide. You're tasked with comparing the populations of capital cities from different countries to identify trends and patterns in urban demographics.

```sql
SELECT country.name AS Country, city.name AS Capital_city, city.population
FROM country
JOIN city ON country.Capital = city.ID
ORDER BY city.Population DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Country	Capital_city	population
South Korea	Seoul	9981619
Indonesia	Jakarta	9604900
Mexico	Ciudad de México	8591309
Russian Federation	Moscow	8389200
Japan	Tokyo	7980230
China	Peking	7472000
United Kingdom	London	7285000
Egypt	Cairo	6789479
Iran	Teheran	6758845
Peru	Lima	6464693
Thailand	Bangkok	6320174
Colombia	Santafé de Bogotá	6260862
Congo, The Demo...	Kinshasa	5064000
Chile	Santiago de Chile	4703954
Iraq	Baghdad	4336000
Singapore	Singapore	4017733
Bangladesh	Dhaka	3612850
Germany	Berlin	3386667
Myanmar	Rangoon (Yangon)	3361700
Saudi Arabia	Riyadh	3324000
Turkey	Ankara	3038159

Result 9 | Read Only

Output

18. **Countries with Low Population Density:** *Scenario:* An agricultural research institute is studying countries with low population densities for potential agricultural development projects. You're tasked with identifying countries with sparse populations from the database to support the institute's research efforts.

```sql
SELECT Name AS Country,
       Population / SurfaceArea AS Population_Density
FROM country
WHERE SurfaceArea > 0
ORDER BY Population_Density ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Country	Population_Density
Antarctica	0.0000
French Southern territories	0.0000
Bouvet Island	0.0000
Heard Island and McDonald Islands	0.0000
British Indian Ocean Territory	0.0000
South Georgia and the South Sandwich Islands	0.0000
United States Minor Outlying Islands	0.0000
Greenland	0.0259
Svalbard and Jan Mayen	0.0513
Falkland Islands	0.1643
Pitcairn	1.0204
Western Sahara	1.1015
Mongolia	1.6993
French Guiana	2.0111
Namibia	2.0939
Australia	2.4397
Suriname	2.5541
Mauritania	2.6036
Iceland	2.7087
Botswana	2.7882
Canada	3.1239

Result 19 | Read Only

19. **Cities with High GDP per Capita:** *Scenario:* An economic consulting firm is analysing cities with high GDP per capita for investment opportunities. You're tasked with identifying cities with above-average GDP per capita from the database to assist the firm in identifying potential investment destinations.

```sql
SELECT city.Name AS City, country.Name AS Country, (country.GNP / country.Population) AS GDP_Per_Capita
FROM city
JOIN country
ON city.CountryCode = country.Code
WHERE country.GNP IS NOT NULL
AND (country.GNP / country.Population) > (SELECT AVG(GNP / Population) FROM country WHERE GNP IS NOT NULL)
ORDER BY GDP_Per_Capita DESC;
```

City	Country	GDP_Per_Capita
Luxembourg [Luxemburg/Lëtzebuerg]	Luxembourg	0.037459
Zürich	Switzerland	0.036936
Geneve	Switzerland	0.036936
Basel	Switzerland	0.036936
Bern	Switzerland	0.036936
Lausanne	Switzerland	0.036936
Saint George	Bermuda	0.035815
Hamilton	Bermuda	0.035815
Bandar Seri Begawan	Brunei	0.035686
Schaan	Liechtenstein	0.034644
Vaduz	Liechtenstein	0.034644
George Town	Cayman Isla...	0.033237
København	Denmark	0.032664
Århus	Denmark	0.032664
Odense	Denmark	0.032664
Aalborg	Denmark	0.032664
Frederiksberg	Denmark	0.032664
Oslo	Norway	0.032577
Bergen	Norway	0.032577
Trondheim	Norway	0.032577
Stavanger	Norway	0.032577
Bærum	Norway	0.032577
New York	United States	0.030575
Los Angeles	United States	0.030575
Chicago	United States	0.030575

20. **Display Columns with Limit (Rows 31-40):** *Scenario:* A market research firm requires detailed information on cities beyond the top rankings for a comprehensive analysis. You're tasked with providing data on cities ranked between 31st and 40th by population to ensure a thorough understanding of urban demographics.

```
SELECT Name, CountryCode, Population
FROM city
ORDER BY Population DESC
LIMIT 10 OFFSET 30;
```

Result Grid

Name	CountryCode	Population
Shenyang	CHN	4265200
Kanton [Guangzhou]	CHN	4256300
Singapore	SGP	4017733
Ho Chi Minh City	VNM	3980000
Chennai (Madras)	IND	3841396
Pusan	KOR	3804522
Los Angeles	USA	3694820
Dhaka	BGD	3612850
Berlin	DEU	3386667
Rangoon (Yangon)	MMR	3361700

Course Notes

It is recommended to take notes from the course, use the space below to do so, or use the revision guide shared with the class:

We have included a range of additional links to further resources and information that you may find useful, these can be found within your revision guide.

END OF WORKBOOK

Please check through your work thoroughly before submitting and update the table of contents if required.

Please send your completed work booklet to your trainer.